Data Fusion: What Drives Student Exam Scores?

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Data fusion combines different sources and types of data to give a fuller picture than one dataset alone (Jagadish F, 2024). I applied it to the Student Performance Factors dataset (lainguyn123, n.d.), which joins behavior data (attendance, study hours), personal data (sleep, prior scores), and home data (parental involvement, resources) for 6,607 students.

Data Exploration and Quality

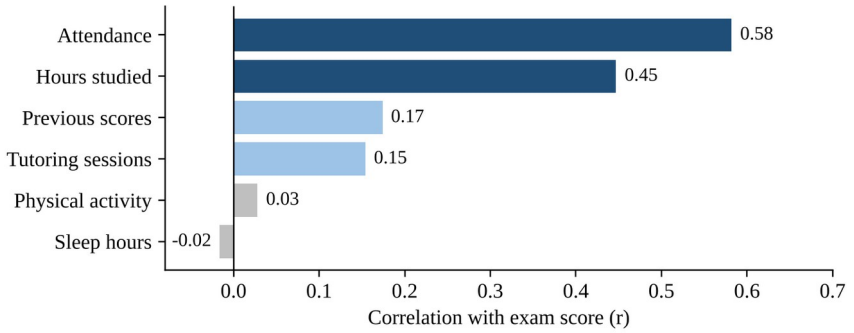
I first checked the data in Excel with COUNTBLANK, MIN, and MAX. There were 235 missing values in three categorical columns and one exam score of 101, which is impossible on a 100-point scale. I removed that record (6,606 remain) and left missing values out only where those columns were used. Checking quality first keeps small errors from quietly skewing every later result.

Correlations

Using CORREL (Figure 1), attendance had the strongest link to exam scores (r = .58), followed by hours studied (r = .45). Sleep and physical activity showed almost no link. Students attending 90% or more averaged 70.1, versus 64.2 below 70%.

Figure 1

Correlation of Each Factor With Exam Score

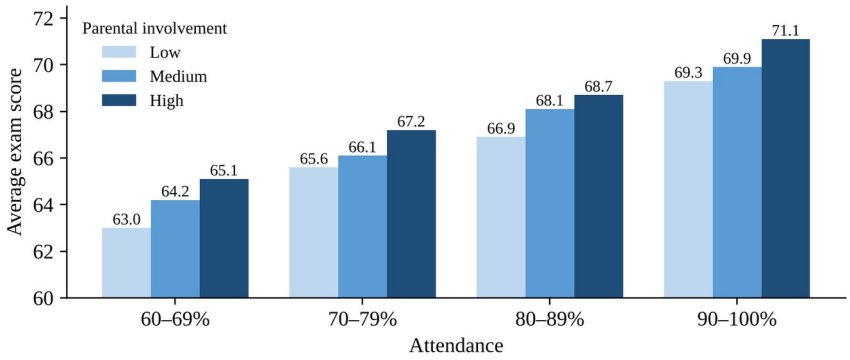


What Combining the Data Adds

Attendance alone explained 34% of score variation and study hours 20%, but together (LINEST) they explained 54%, because the two are unrelated ($r = -.01$) and each adds new information. Home data added another layer: at every attendance level, high parental involvement meant about 2 more points (Figure 2 and the pivot table). Students with 90%+ attendance and over 20 study hours averaged 71.7, almost 9 points above those with under 70% and 20 hours or less (62.9).

Figure 2

Average Exam Score by Attendance and Parental Involvement



Correlation or Causation, and Recommendations

These are correlations, not proof of cause (Black, 2019). Attendance was about 80% at every motivation level, so motivation alone does not explain the link, but the data is observational. A school should flag students below 80% attendance early, encourage 20+ study hours a week, keep parents involved, and test these steps in a small pilot against a comparison group before scaling up. Checking quality, combining data, and questioning cause in this order reduces uncertainty before acting.

Use of AI: Claude (Anthropic) helped clean and explore the data, build the Excel formulas, pivot table, and charts, and draft and edit this report. I reviewed all results against the data.

References

Black, K. (2019). Simple regression analysis and correlation. In J. W. Foreman, *Foundations of data analytics* (1st ed., Module 5). Wiley.

Jagadish F. (2024, February 3). *What is data fusion in data analytics?* Fabric. https://blog.fabrichq.ai/what-is-data-fusion-in-data-analytics-1fe0698c3a53

lainguyn123. (n.d.). *Student performance factors* [Data set]. Kaggle. https://www.kaggle.com/datasets/lainguyn123/student-performance-factors